An Auditable Clinical Data Warehouse for Time-Bounded ICU Research

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Technical manuscript — 18 October 2026

Abstract

This paper describes a MIMIC-IV-shaped clinical warehouse and a constrained prospective study workbench. Its contribution is an executable chain from source provenance to cohort construction, measurement availability, model evaluation, and aggregate publication. DuckDB Bronze, Silver, and Gold layers support local analytics; a versioned study contract defines first-ever ICU stay, prediction at 24 hours, and death after prediction before hospital discharge. A synthetic generator enables end-to-end execution without credentialed MIMIC-IV records. In one identified synthetic run, 560/120/120 patients entered training/validation/test partitions, with test average precision 0.109 and AUROC 0.535. These values test software behavior, not clinical validity. We specify the temporal predicates, preprocessing and split protocol, controlled quality perturbations, and current provenance and disclosure-control gaps. The resulting platform is a basis for research on temporal validity and transportability once approved real-data access and stronger reproducibility controls are available.

Keywords: MIMIC-IV; EHR; clinical data warehouse; temporal leakage; prediction; synthetic data.

1 Motivation and research boundary

MIMIC-IV combines hospital and ICU records useful for secondary research [1]. Full v3.1 files require credentialing, training, and a data-use agreement [2]; the open clinical demo is a separate, small resource [3]. This repository contains neither an authorized full-data extract nor a model trained on one. Its local results use invented records. That boundary matters because an internally consistent synthetic score cannot estimate discrimination, calibration, fairness, or utility in patients.

The research question is infrastructural: *can a study's source coverage, index stay, prediction time, eligible observations, split, and published evidence be reconstructed from executable artifacts?* This focus follows two known risks. EHR measurements reflect care processes as well as physiology [4]; and prediction studies need clear reporting of participants, predictors, outcome, analysis, and limitations [5, 6]. We therefore treat the cohort and time contract as first-class study objects rather than relying on a final model score.

2 Warehouse architecture

The system accepts source-shaped CSV/CSV.GZ files through local ingestion, a small invented demo seed, an isolated 1,000-patient research generator, or a configured LocalStack S3 path. BigQuery support is metadata-only. The 1,000-patient generator writes ignored source files and a separate DuckDB database. Neither its distributions nor its privacy properties were learned from restricted MIMIC-IV data.

*Repository: https://github.com/pronabroy360/MIMIC-IV-Clinical-DW.

Bronze tables retain source-like fields plus `source_file`, `loaded_at`, `source_module`, and `pipeline_run_id`. Silver standardizes patient, admission, stay, laboratory, chart, and intervention entities, including event and storage timestamps. Gold builds descriptive patient, admission, ICU, laboratory, and first-24-hour marts. Missing optional inputs create empty placeholders to keep the pipeline executable; source coverage must therefore be inspected before interpreting an empty mart as a true zero. A distinct research feature table prevents whole-stay descriptive summaries from entering a 24-hour prediction task.

The Makefile and CLI orchestrate ingestion, transformation, validation, and reporting. FastAPI and Streamlit expose aggregate analytics and call a shared study service. The rule-based query assistant supports predefined aggregate prompts; SQL guardrails reject destructive statements, Bronze access, raw identifiers, and nonaggregate projections. These checks are application controls, not a formal privacy proof or production authorization system.

Layer	Stored object	Inspection point
Bronze	Source-shaped rows and ingest metadata	File coverage and provenance
Silver	Typed entities, keys, event times, storage times	Join, uniqueness, and date checks
Gold	Descriptive marts	Aggregate counts and dashboards
Study	Eligible stays, timed features, model output	Cohort flow, feature/timing audits
Release	Allowlisted aggregate package	Manifest and SHA-256 checksums

Table 1: Evidence chain implemented by the local warehouse and study service.

3 Prospective ICU study contract

3.1 Index population and estimand

The versioned YAML contract fixes synthetic origin, task and feature versions, seed 42, 70/15/15 patient split, and a 24-hour horizon. SQL chooses each patient's first-ever ICU stay by `intime`, breaking ties with `stay_id`, *before* applying adult, optional care-unit, and eligibility filters. Consequently, a later eligible stay cannot replace an excluded first stay. Let t_0 be first ICU entry and $t_p = t_0 + 24\,\text{h}$. Eligibility requires age ≥ 18, known t_0, ICU departure $> t_p$, and no recorded death at or before t_p. The binary outcome is

$$Y = \mathbf{1}\{t_p < t_{\text{death}} \leq t_{\text{hospital discharge}}\}. \tag{1}$$

Thus the target population comprises adults alive and still in ICU at 24 hours of their first stay. It excludes early deaths and early ICU departures and says nothing about death after discharge. The preview reports cumulative stay and patient counts from all ICU stays to eligible rows, making selection loss visible.

3.2 Event-time and availability-time predicates

For observation event time t_e and storage time t_s, the research feature SQL enforces

$$t_0 \leq t_e < t_p, \qquad t_e \leq t_s \leq t_p, \qquad t_s \neq \text{NULL}. \tag{2}$$

The half-open event interval excludes a measurement exactly at prediction. A result recorded after prediction is unavailable even if its specimen or chart time is earlier. MIMIC-IV documentation describes laboratory `charttime` as usually near specimen acquisition and `storetime` as result availability; ICU chart storage denotes entry or validation rather than necessarily the instant of bedside observation [7, 8]. Accordingly, `storetime` is an operational availability proxy whose meaning must be checked by table and workflow before a real-data study.

Chart and input events join the selected `stay_id`; hospital labs join `hadm_id` and must also fall inside the stay's ICU window. The builder aggregates six vital-sign families, nine laboratory families, event counts, and a first-24-hour vasopressor flag. Age and three categorical context fields complete a task registry of 19 numeric and three categorical predictors. Diagnosis/procedure codes, discharge fields, identifiers, site label, future events, and whole-stay summaries are excluded. A feature audit records coverage, missing rate, distinct values, and activity status. This is an audit, not automatic variable selection.

4 Evaluation and controlled perturbations

4.1 Internal evaluation

The flagship task uses one selected stay per patient. Patient identifiers determine a seeded 70/15/15 train/validation/test assignment. A scikit-learn pipeline fits median imputation and standardization for numeric variables, most-frequent imputation and one-hot encoding for categorical variables, and class-weighted logistic regression on training data. Fitting preprocessing inside the pipeline prevents validation/test statistics from entering training. Validation average precision is reported; a separate locked test partition supplies final metrics. Repeated stratified three-fold cross-validation, twice on train plus validation, describes development stability rather than an independent post-selection estimate. The current flagship fits one logistic candidate; its artifact named `model_comparison.csv` must not be read as evidence of a broad algorithm search.

The output reports AUROC, average precision (AP), and Brier score. Precision–recall analysis is informative for imbalanced outcomes [9]; probability error is summarized as $n^{-1}\sum_{i=1}^{n}(p_i-y_i)^2$. A Brier score alone does not establish calibration; a future real-data protocol should add calibration intercept, slope, and plots [10]. Test uncertainty uses 200 patient-row bootstrap resamples and 2.5/97.5 percentiles when two classes are present. Signed, baseline-centered linear contributions are checked for additivity; they are model-scale explanations, not causal effects or SHAP output.

Published synthetic run `...970d76be`	Value
Train / validation / test rows	560 / 120 / 120
Development CV AP, mean (SD)	0.120 (0.027)
Validation AP	0.130
Test AUROC / AP / Brier	0.535 / 0.109 / 0.241

Table 2: Rounded values from the identified aggregate run package; manifest revision `e8fdf87`. They quantify an invented-data exercise only.

Another published package with the same definition hash has different test scores. The present `dataset_snapshot` hashes only Silver table *names*, not row contents; the manifest records Git HEAD but no full uncommitted patch or locked environment. Therefore, those packages do not prove identical inputs or code. Content hashes, a persisted private split manifest, frozen dependencies, and complete code-state capture are required before claiming bitwise reproducibility.

4.2 Quality and site experiments

Isolated invented scenarios delay laboratory storage time, duplicate admissions, remove ICU anchors, or introduce impossible oxygen saturation. The paired comparison fixes a baseline model and compares common held-out patients while reporting attrition separately; quality-failing scenarios are intended to block score publication. This distinguishes changing inputs from changing the cohort. A two-site exercise simulates delayed labs at site B and fits recalibration on a disjoint site-B partition. Because both sites originate from one generator, it demonstrates a workflow rather than geographic external validity. Genuine transportability needs independent institutions, code/unit mapping, and prespecified external validation [11].

5 Release controls, limits, and research programme

The publisher copies an explicit allowlist of aggregate files: flow, timing and feature audits; model and evaluation summaries; signed attributions; manifest; quality summary; and README. It writes SHA-256 checksums. Patient-level predictions and split assignments stay outside that package. A minimum cell count exists in the YAML contract, but the copier does not enforce it on every field. Restricted-data release needs an artifact-level disclosure review. SQL guardrails and ignored raw paths reduce accidental exposure without establishing a comprehensive security guarantee.

The earlier 1,000-patient teaching workshop remains distinct: its first-admission grain supports length of stay, in-hospital death, urgent 30-day readmission, and a synthetic diagnosis class with a 73-candidate catalog. Some fields have retrospective availability, and the diagnosis target is not a clinical phenotype. Neither workshop scores nor the prospective synthetic run demonstrate performance on MIMIC-IV. Empty placeholder tables, simple mean features, incomplete content fingerprinting, and no independent-site test further limit interpretation. TRIPOD+AI and PROBAST offer reporting and bias-assessment structures for a future clinical study, but this prototype does not claim compliance or low risk of bias [5, 6].

The next study should proceed in stages: (1) hash input contents, SQL, code, environment, and split assignments; (2) audit each field's clinical availability and missingness mechanism; (3) execute a prespecified cohort and analysis plan inside an approved MIMIC-IV environment; (4) report discrimination, calibration, uncertainty, and subgroup performance; and (5) test temporal and independent-site transportability before discussing clinical use. Those are research objectives, not results of the current repository.

6 Conclusion

The platform's present achievement is an inspectable, executable study pathway: source coverage, first-stay selection, two-clock feature eligibility, patient separation, aggregate evaluation, and bounded publication can be traced together. Synthetic scores establish that the software runs under an explicit contract. Clinical claims await credentialed data, verified availability semantics, stronger provenance, and independent validation.

Reproducibility note

The study YAML is under `configs/studies/`; service code is in `src/research/`. Generate data with `make research-demo`. Then run `make study-validate`, `make study-preview`, and `make study-run`. Table 2 refers to an earlier published package; the current Git head does not guarantee its scores.

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